

NBA vs Euroleague Metrics Reference Guide

Why Use European Metrics?

When analyzing Euroleague basketball, using native European measurements (metric system) provides several advantages:

- 1. **Native Data Format:** Euroleague API returns data in metric units
- 2. **Court Accuracy:** Actual Euroleague court dimensions are metric
- 3. **No Conversion Errors:** Eliminates rounding errors from unit conversion
- 4. **International Standard:** Easier to compare with other European leagues
- 5. **Data Integrity:** Preserves original measurement precision

Court Dimensions Comparison

Measurement	NBA (Imperial)	NBA (Metric)	Euroleague (Metric)
Court Length	94 feet	28.65 m	28 m
Court Width	50 feet	15.24 m	15 m
3-Point Line (Top)	23.75 feet	7.24 m	6.75 m
3-Point Line (Corner)	22 feet	6.71 m	6.75 m
Free Throw Line	15 feet	4.57 m	4.60 m
Basket Height	10 feet	3.05 m	3.05 m
Paint Width	16 feet	4.88 m	4.90 m

Key Insight: Euroleague has a **shorter 3-point line** (6.75m vs 7.24m), which affects shot distribution analysis.

Shot Zone Definitions

NBA Zones (Imperial → Metric)

Zone Name	Imperial	Metric Conversion
At Rim	0-3 feet	0-0.91 m

Zone Name	Imperial	Metric Conversion
Short Midrange	3-10 feet	0.91-3.05 m
Long Midrange	10 ft - 3P	3.05 m - 7.24 m
Three-Point	23.75+ feet	7.24+ m

Euroleague Zones (Native Metric)

Zone Name	Metric (Native)	Description
At Rim	0-0.91 m	Paint area, dunks, layups
Short Midrange	0.91-3.05 m	Floaters, short jumpers
Long Midrange	3.05-6.75 m	Mid-range to 3-point line
Three-Point	6.75+ m	Beyond the arc
Corner Three	6.75 m (corner)	Same distance as top of key

Advantage: Euroleague corner threes are the **same distance** as top-of-key threes (6.75m), unlike NBA where corners are shorter.

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Player Height Reference

NBA Convention

- Typically listed in **feet and inches** (e.g., 6'8")
- Sometimes converted to inches (80 inches)

Euroleague Convention

- Always listed in **centimeters** (e.g., 203 cm)
- More precise measurements
- No fractional inches

Quick Conversion Reference

Position	NBA (ft'in")	Metric (cm)
Point Guard	6'2"	188 cm

Position	NBA (ft'in ")	Metric (cm)
Shooting Guard	6'5"	196 cm
Small Forward	6'7"	201 cm
Power Forward	6'9"	206 cm
Center	6'11"	211 cm

Statistical Variables: NBA vs Euroleague Format

Distance-Based Variables

Variable	NBA Format	Euroleague Format	Conversion
0-3 ft FGA%	Feet	0-0.91m FGA %	×0.3048
3-10 ft FGA%	Feet	0.91-3.05m FGA %	×0.3048
10ft-3p FGA%	Feet	3.05m-3p FGA %	×0.3048

Rate-Based Variables (Same Format)

These remain the same regardless of measurement system:

- Offensive Rebound Rate (%)
- Defensive Rebound Rate (%)
- Assist Rate
- Steal Rate (per game)
- Block Rate (per game)
- Turnover Rate (per 100)
- Points Per 100 Possessions
- Usage Rate (%)
- Free Throw Rate
- 3-Point Attempt Rate (%)

Practical Example: Shot Distance Analysis

NBA Approach (Imperial)

```
python

# NBA: Convert everything to feet
court_length_ft = 94
three_point_ft = 23.75

if shot_distance_ft <= 3:
    zone = "rim"
elif shot_distance_ft <= 10:
    zone = "short_mid"
```

Euroleague Approach (Metric) ✓

```
python

# Euroleague: Use native meters
court_length_m = 28
three_point_m = 6.75

if shot_distance_m <= 0.91:
    zone = "rim" # 0-0.91m
elif shot_distance_m <= 3.05:
    zone = "short_mid" # 0.91-3.05m
elif shot_distance_m < 6.75:
    zone = "long_mid" # 3.05m-6.75m
else:
    zone = "three" # 6.75m+
```

No conversion needed! Data flows naturally from API → Analysis → Results.

Data Collection Best Practices

✓ DO: Keep Native Metrics

```
python

# Good: Preserve original units
player_height_cm = 203
shot_distance_m = 2.45
court_width_m = 15.0
```

✗ DON'T: Unnecessary Conversions

```
python

# Bad: Creates rounding errors
player_height_in = 203 / 2.54 # 79.92 inches (lost precision)
shot_distance_ft = 2.45 * 3.28084 # 8.04 feet (unnecessary)
```

Variable Names Convention

We use clear suffixes to indicate measurement units:

Suffix	Meaning	Example
<code>_cm</code>	Centimeters	<code>Height_cm = 203</code>
<code>_m</code>	Meters	<code>0-0.91m_FGA_Pct</code>
<code>_Pct</code>	Percentage	<code>3FGA_Pct = 35.2</code>
<code>_Rate</code>	Rate/Ratio	<code>OReb_Rate = 0.15</code>
<code>_Per100</code>	Per 100 possessions	<code>Points_Per100 = 18.5</code>

Impact on Clustering Analysis

Using native Euroleague metrics ensures:

- Accurate Player Comparisons:** Heights and distances reflect actual measurements
- Proper Shot Zone Distribution:** Zones match actual court layout
- Cross-Season Consistency:** No conversion drift over time
- International Compatibility:** Easy to add other European leagues

Example cluster interpretation becomes more natural:

- "Players who shoot 45% from 0-0.91m" (clear metric distance)
- "Centers averaging 205cm in height" (actual Euroleague measurement)
- "Guards taking 40% of shots beyond 6.75m" (actual 3-point line)

Summary: Key Metric Values to Remember

Measurement	Value	Context
6.75 m	Three-point line	Shorter than NBA (7.24m)
0.91 m	Rim zone boundary	~3 feet, at-rim area
3.05 m	Mid-range split	Separates short/long midrange
28 m × 15 m	Court dimensions	Slightly smaller than NBA
203 cm	Average center height	~6'8", typical Euroleague big
188 cm	Average guard height	~6'2", typical Euroleague guard

Recommendation: Always work in native European metrics when analyzing Euroleague data. Only convert to imperial if specifically required for cross-league comparisons with NBA.